Here's a **structured learning roadmap** tailored to your **career goal as a Cloud and DevOps Engineer**, starting from basics to advanced, including hands-on project experience and certification goals.

Phase 1: Foundations (1-2 months)

 Goal: Understand basics of Linux, networking, and Git

- **Linux & Shell Scripting**
  - Basic commands, file system, permissions
  - Writing and executing shell scripts
  - Tools: vim, grep, awk, sed, crontab
- **Networking Basics**
  - IP, DNS, HTTP/HTTPS, firewalls, ports
  - NAT, subnets, load balancing
- **Git & GitHub**
  - Version control basics
  - Branching, merging, pull requests
  - GitHub workflows and repositories

 Resources:

- Linux Journey
- Pro Git Book

 Mini Project: Host a static website using GitHub Pages

Phase 2: DevOps Essentials (2-3 months)

 Goal: Automate code integration, testing, deployment

- **CI/CD Tools**
  - Jenkins (install, pipelines, freestyle jobs)
  - GitLab CI, GitHub Actions, Azure DevOps, AWS CodePipeline
- **Build Tools**
  - Maven, Gradle, or Node-based builds (depending on app type)
- **Artifact Management**
  - JFrog Artifactory or GitHub Packages
- **Containerization**
  - Docker basics: build, run, tag, push to Docker Hub
  - Docker Compose

 Resources:

- Jenkins Docs

- Docker's Getting Started

 Project: Create a CI/CD pipeline for a sample Spring Boot or Node.js app using Jenkins and Docker

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Phase 3: Cloud Platform (3-4 months)

 Goal: Learn and deploy apps using AWS, Azure, or GCP

- **Choose 1 Cloud Platform (start with AWS)**
  - EC2, S3, IAM, RDS, Lambda, VPC, CloudWatch
  - CLI, SDK, and Console
  - Deploy static and dynamic apps
- **Infrastructure as Code (IaC)**
  - Terraform: variables, modules, state
  - CloudFormation (for AWS)

 Resources:

- AWS Free Tier Labs
- Terraform Docs

 Projects:

- Deploy a Dockerized app to EC2 using Terraform
- Serverless REST API with Lambda + API Gateway

 Certification:

- **AWS Certified Solutions Architect – Associate** (or Azure/GCP equivalent)

Phase 4: Container Orchestration & Monitoring (2-3 months)

 Goal: Scale and manage microservices effectively

- **Kubernetes**
  - Architecture, Pods, Deployments, Services
  - Helm charts
  - EKS (AWS), AKS (Azure), or GKE (GCP)
- **Monitoring & Logging**
  - Prometheus + Grafana
  - ELK Stack (Elasticsearch, Logstash, Kibana)
  - Cloud-native monitoring (CloudWatch, Azure Monitor, etc.)

 Resources:

- Kubernetes Docs
- Play with K8s

 Project: Deploy a full-stack app (frontend, backend, DB) on Kubernetes with Helm and monitoring

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Phase 5: Advanced DevOps Practices (2-3 months)

 Goal: Build real-world scalable infrastructure

- **Configuration Management**
  - Ansible or Chef: provisioning, templating, roles
- **Security & Cost Optimization**
  - IAM roles, secrets management (AWS Secrets Manager, Vault)
  - Budgets, cost alerts
- **DevSecOps & GitOps**
  - Integrate security into pipelines
  - GitOps tools: ArgoCD or Flux
- **Site Reliability Engineering (SRE) Basics**
  - SLIs, SLOs, Error Budgets, Incident Response

 Resources:

- Google SRE Book

 Capstone Project:

- CI/CD pipeline + IaC + Kubernetes + Monitoring + Secrets + Auto-scaling

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Bonus: Soft Skills & Career Growth

- Write technical blogs on Dev.to or Medium
- Contribute to open-source on GitHub
- Prepare for behavioral & system design interviews
- Build a strong LinkedIn & GitHub portfolio

 Summary Timeline

Phase	Duration	Outcome
Foundations	1-2 months	Shell, Git, Networking
DevOps Basics	2-3 months	CI/CD, Docker
Cloud Platform	3-4 months	AWS/Azure/GCP, Terraform
Containers & Monitoring	2-3 months	Kubernetes, Prometheus, ELK
Advanced Practices	2-3 months	Ansible, GitOps, SRE, Security

Would you like a **customized roadmap PDF**, Notion board, or GitHub-based checklist for tracking your progress?

Great! Here's how we'll proceed:

I'll give you:

1. A **Notion Board Template** link (for task tracking & notes)
2. A **GitHub-based Checklist (Markdown)** you can paste into your repo's README

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 1. Notion Board Template – DevOps & Cloud Engineer Roadmap

 Notion Template Link:
 Click to Duplicate the Board

 What's Included:

- Columns for each phase (To Learn, In Progress, Done)
- Tasks for each topic with links
- Note section for key learnings or bookmarks
- Certification tracker

Click **"Duplicate"** in the top-right to copy it into your Notion workspace

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 2. GitHub-Based Checklist (Markdown Format)

You can copy-paste this into your README.md or progress.md in your GitHub repo:

DevOps & Cloud Engineer Learning Checklist

□ Phase 1: Foundations
- [ ] Learn Linux basics (commands, file system, permissions)
- [ ] Practice shell scripting
- [ ] Understand networking (IP, DNS, Ports, etc.)
- [ ] Learn Git (clone, commit, push, branching)
- [ ] Create and host static website on GitHub Pages

□ Phase 2: DevOps Essentials
- [ ] Install and use Jenkins
- [ ] Create basic CI/CD pipeline (build, test, deploy)
- [ ] Learn GitHub Actions / GitLab CI
- [ ] Use Docker: build, tag, run, push images
- [ ] Deploy Docker app to local or cloud

☁ Phase 3: Cloud Platform (AWS Preferred)
- [ ] Learn EC2, S3, IAM, RDS, Lambda, VPC
- [ ] Deploy static website to S3
- [ ] Use AWS CLI & Management Console
- [ ] Learn Infrastructure as Code with Terraform
- [ ] Deploy EC2 + S3 via Terraform

□ Phase 4: Containers & Monitoring
- [ ] Learn Kubernetes basics: Pods, Deployments, Services
- [ ] Use Helm for package management
- [ ] Deploy app to Kubernetes cluster (Minikube or EKS)
- [ ] Monitor with Prometheus + Grafana
- [ ] Implement ELK stack for logging

□ Phase 5: Advanced DevOps
- [ ] Use Ansible for provisioning
- [ ] Learn secrets management (AWS Secrets Manager or Vault)
- [ ] Setup GitOps with ArgoCD or Flux
- [ ] Study DevSecOps (SAST, DAST, policy enforcement)
- [ ] Understand SRE basics (SLI, SLO, alerts)

□ Capstone Projects
- [ ] CI/CD + Docker + Terraform + Kubernetes full-stack project
- [ ] Add monitoring & alerting
- [ ] Publish your GitHub project with README and architecture diagram

□ Certification Goals
- [ ] AWS Certified Solutions Architect Associate
- [ ] Certified Kubernetes Administrator (CKA)
- [ ] Terraform Associate (optional)

□ Ongoing

- [ ] Share blogs on Dev.to or Medium
- [ ] Network on LinkedIn
- [ ] Prepare for interviews (behavioral + system design)